Aquarium After Hours

Whodunnit? · Graphing Inequalities · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Graphing Inequalities	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Writing the linear inequality that matches a graph (line type and shading).

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Prof. Tide took it, in the Penguin Cove, using this method: Stuffed in a plush octopus.

The reveal (read aloud): While the jellyfish glowed, **Professor Tide** slipped into **Penguin Cove**, tucked the Pearl of the Deep Blue into a **plush octopus**, and strolled out hugging it like a souvenir. Every clue you cleared pointed to him. Dive-in great work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Captain Reyes · Place: Dive Deck
2	Method: Hidden in a fish bucket
3	Suspect: Dr. Kelp · Place: Touch Pool
4	Method: Smuggled in a dive bag
5	Suspect: Ms. Nori · Place: Gift Shop
6	Method: Tucked in a stroller
7	Suspect: Mr. Finley · Place: Shark Tunnel
8	Method: Wrapped in a wetsuit
9	Suspect: Coral Chen · Place: Jellyfish Gallery
10	Method: Disguised as a snow globe

Answer Key

Aquarium After Hours · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	A **dashed** boundary line passes through (0, 1) and (1, 3). The region **above** the line is shaded.	**A.** $y > 2x + 1$	slope = 2, y-intercept = 1 dashed → >, above → >$y > 2x + 1$	Captain Reyes was checking tanks on the Dive Deck. The equipment log has his initials.
2	A **solid** boundary line passes through (0, 1) and (1, 3). The region **below** the line is shaded.	**B.** $y \le 2x + 1$	slope = 2, y-intercept = 1 solid → ≤, below → ≤$y \le 2x + 1$	The fish buckets were all emptied and rinsed. No pearl inside.
3	A **dashed** boundary line passes through (0, 1) and (1, −1). The region **below** the line is shaded.	**B.** $y < -2x + 1$	slope = −2, y-intercept = 1 $y < -2x + 1$	Dr. Kelp was showing kids a starfish at the Touch Pool.
4	A **solid** boundary line passes through (0, 1) and (−1, 3). The region **above** the line is shaded.	**D.** $y \ge -2x + 1$	slope = $\frac{3-1}{-1-0}$ = −2, y-intercept = 1 $y \ge -2x + 1$	Every dive bag was searched at the exit. None held the Pearl.
5	A **dashed** boundary line has slope 2 and y-intercept −1. The region **above** the line is shaded.	**D.** $y > 2x - 1$	dashed, above → >$y > 2x - 1$	Ms. Nori was ringing up souvenirs in the Gift Shop. The receipts are time-stamped.
6	A **solid** boundary line has slope −2 and y-intercept −1. The region **below** the line is shaded.	**C.** $y \le -2x - 1$	solid, below → ≤ $y \le -2x - 1$	The strollers were checked at the door. Only snacks and blankets.
7	A **dashed** boundary line passes through (0, −1) and (2, 0). The point (0, 0) is in the shaded region.	**D.** $y > \frac{1}{2}x - 1$	slope = $\frac{1}{2}$ (0, 0) is above the line (0 > −1), so shade above $y > \frac{1}{2}x - 1$	Mr. Finley was feeding the sharks in the Shark Tunnel while a tour group watched.
8	A **solid** boundary line passes through (0, −1) and (−2, −2). The shaded region is **below** the line.	**C.** $y \le \frac{1}{2}x - 1$	slope = $\frac{-2-(-1)}{-2-0}$ = $\frac{1}{2}$ $y \le \frac{1}{2}x - 1$	The wetsuits on the rack are all dry and flat. Nothing was wrapped in one.
9	A **dashed** boundary line passes through (0, 2) and (2, 1). The point (0, 0) is in the shaded region.	**A.** $y < -\frac{1}{2}x + 2$	slope = $-\frac{1}{2}$ (0, 0) is below the line (0 < 2), so shade below $y < -\frac{1}{2}x + 2$	Coral Chen was leading a tour through the Jellyfish Gallery.
10	A **solid** boundary line passes through (0, 2) and (−2, 3). The shaded region is **above** the line.	**A.** $y \ge -\frac{1}{2}x + 2$	slope = $\frac{3-2}{-2-0}$ = $-\frac{1}{2}$ $y \ge -\frac{1}{2}x + 2$	The snow globes on the shelf are sealed and unopened. The Pearl isn't in any of them.

Wrong-Answer Map

Aquarium After Hours · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	B. $y < 2x + 1$	shaded wrong side	Captain Reyes was posing for a photo with the penguins in Penguin Cove. The camera clock says 9:04 p.m.
1	C. $y \geq 2x + 1$	solid line instead of dashed	Prof. Tide was measuring water samples in the Shark Tunnel. The lab notebook has a 9:02 p.m. entry.
1	D. $y > \frac{1}{2}x + 1$	inverted slope	Every plush octopus in the Gift Shop was unzipped and checked. Only stuffing inside.
2	A. $y < 2x + 1$	dashed line instead of solid	Prof. Tide was showing a sea star to a family at the Touch Pool. They took a photo together.
2	C. $y \geq 2x + 1$	shaded wrong side	Dr. Kelp was checking penguin eggs in Penguin Cove. The nest chart has his handwriting.
2	D. $y \leq \frac{1}{2}x + 1$	inverted slope	The plush octopuses on the shelf are all still sealed in plastic. None held the Pearl.
3	A. $y > -2x + 1$	shaded wrong side	Ms. Nori was handing out penguin stickers in Penguin Cove. A kid still has one on his shirt.
3	C. $y \leq -2x + 1$	solid instead of dashed	Prof. Tide was giving a short talk in the Jellyfish Gallery. The whole audience stayed seated.
3	D. $y < 2x + 1$	sign error in slope	A staff member squeezed every plush octopus. All were soft and squishy all the way through.
4	A. $y > -2x + 1$	dashed instead of solid	Prof. Tide was buying a postcard in the Gift Shop. The receipt is time-stamped 9:03 p.m.
4	B. $y \leq -2x + 1$	shaded wrong side	Mr. Finley was tossing fish to the penguins in Penguin Cove while a crowd laughed.
4	C. $y \geq 2x + 1$	sign error in slope	The plush octopus bin was counted before and after the gala. The numbers match.
5	A. $y > 2x + 1$	swapped sign of intercept	The giant plush octopus by the door was weighed and found to be only fluff.
5	B. $y < 2x - 1$	shaded wrong side	Prof. Tide was helping unload gear on the Dive Deck. The deckhand’s log lists his name.
5	C. $y \geq 2x - 1$	solid instead of dashed	Coral Chen was leading a tour group through Penguin Cove. The group counted heads at 9:05 p.m.
6	A. $y < -2x - 1$	dashed instead of solid	A volunteer saw Prof. Tide feeding the crabs at the Touch Pool when the lights went out.
6	B. $y \geq -2x - 1$	shaded wrong side	The feeding schedule shows Mr. Finley in Penguin Cove from 8:55 to 9:15 p.m.
6	D. $y \leq -2x + 1$	sign error in intercept	The seams on every plush octopus are factory-sewn and intact.
7	A. $y > 2x - 1$	inverted slope	Dr. Kelp was taking the water temperature in Penguin Cove. The reading was logged at 9:00 p.m.
7	B. $y \geq \frac{1}{2}x - 1$	solid instead of dashed	A guest bought a plush octopus at 9:20 p.m. and cut it open at the exit. Just stuffing.

7	C. $y < \frac{1}{2}x - 1$	shaded wrong side	Prof. Tide was pointing out a hammerhead to students inside the Shark Tunnel.
8	A. $y < \frac{1}{2}x - 1$	dashed instead of solid	The Gift Shop owner shook every plush octopus and heard nothing rattle inside.
8	B. $y \le 2x - 1$	inverted slope	Captain Reyes was answering kids' questions at Penguin Cove. A school group backs him up.
8	D. $y \ge \frac{1}{2}x - 1$	shaded wrong side	Prof. Tide was taking notes on moon jellies in the Jellyfish Gallery. His pen ran out of ink there.
9	B. $y \le -\frac{1}{2}x + 2$	solid instead of dashed	Coral Chen was counting penguins in Penguin Cove for her tour script. Every one was present.
9	C. $y > -\frac{1}{2}x + 2$	shaded wrong side	Prof. Tide was coiling a rope on the Dive Deck. A diver says he never left the rail.
9	D. $y < -2x + 2$	inverted slope	Security scanned the plush octopuses with a handheld wand and found nothing hidden.
10	B. $y \ge \frac{1}{2}x + 2$	sign error in slope	Ms. Nori was taking a break on the bench at Penguin Cove. The security camera caught her there.
10	C. $y > -\frac{1}{2}x + 2$	dashed instead of solid	The plush octopus display was photographed at 8:50 and 9:30 p.m., and every octopus looks the same in both.
10	D. $y \le -\frac{1}{2}x + 2$	shaded wrong side	Prof. Tide was signing a book for a fan in the Gift Shop. The fan still has the autograph.